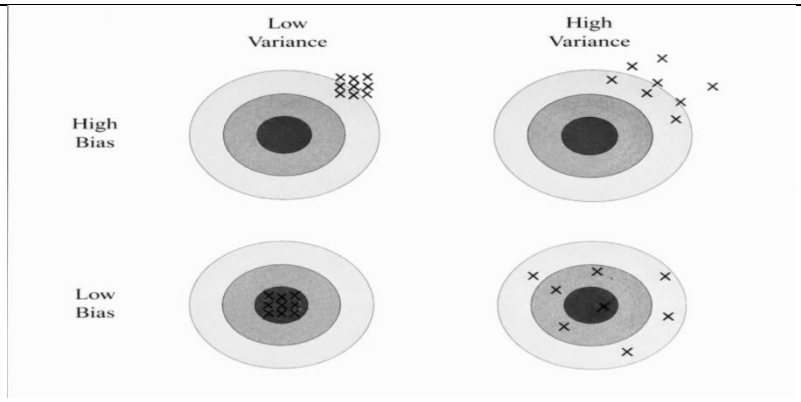
Department	**Department of AI & DS**		
Subject Name	**Machine Learning**	**Subject Code**	**AD-502**

Type of Question- 3 Marks

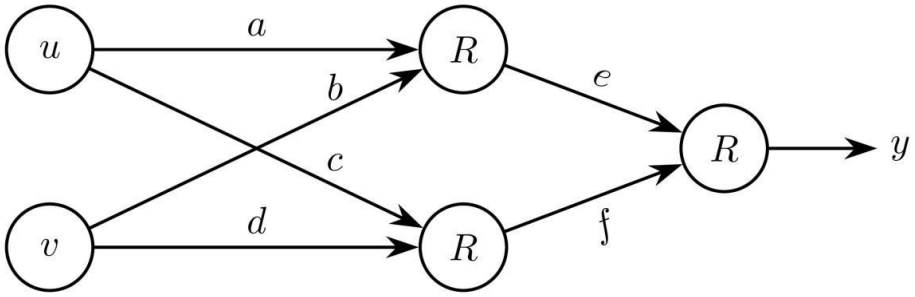
Q.No	**Question**	**Blooms Level**	**CO**
1	Define supervised learning and give examples of tasks that can be solved using this approach.	L2	1
2	Explain various types of machine learning	L2	1
3	Explain the differences and similarities between Artificial Intelligence, Machine Learning, and Deep Learning.	L2	1
4	Carefully read this line and justify this with concepts of Data Preprocessing "AI courses focused on toy datasets with clean values (e.g. UCI census, Kagggle Datasets), but AI in practice required the creation of data pipelines, often from scratch from ground truth to model maintenance".	L2	1
5	Explain the concept of clustering in machine learning. How does it differ from other tasks like classification and regression?	L2	2
6	Given the following **unsorted data for monthly expenses (in dollars)**: 22, 5, 18, 30, 7, 15, 9, 20, 12 1. **Sort the data in ascending order** 2. **Partition the data into equal-frequency bins (3 bins)** 3. **Perform smoothing by bin means** 4. **Perform smoothing by bin boundaries**	L3	1
7	Describe decision tree classification and how it constructs a tree-based model for classification tasks.	L2	3
8	Identify the type of Machine Learning for the following examples a. Finger print recognition of a particular person used in biometric attendance from the finger print data of various other people and that particular person. and explain	L3	1
9	Explain the concept of clustering in machine learning. How does it differ from other tasks like classification and regression?	L2	2
10	Differentiate between Batch and Online Learning	L2	1
11	Differentiate Instance Based Vs Model Based Learning	L2	1
12	Explain the concept of logistic regression in classification. How does it model the probability of class membership?	L2	3
13	Define principal components in the context of PCA. How are they calculated from the original data?	L2	5

14	 A machine learning model is developed to predict **daily energy demand**. The following observations are made: • The model performs poorly on both the **training and test datasets**. • It shows **similar errors across different training datasets**. Answer the following: 1. Based on the **dartboard figure**, which of the four scenarios best represents this model? Justify your answer. 2. Is the model suffering more from **high bias or high variance**? Explain. 3. What is likely causing this behavior in the model	L2	1
15	What is a Gaussian Mixture Model (GMM)? Define the role of a mixture component in GMM.	L2	2
16	Explain the basic idea of fuzzy clustering. How does it differ from hard clustering?	L2	2
17	A Gaussian distribution has mean 10 and standard deviation 2. Calculate the probability density function value at (x=10), using the Gaussian probability density formula.	L3	2
18	Given the following confusion matrix values: TP = 40, TN = 30, FP = 10 and FN = 20, calculate the classification accuracy.	L3	3
19	A K-NN classifier uses K = 3. For a new data point, the three nearest neighbours belong to classes A, A and B. Determine the predicted class using majority voting.	L3	3
20	A classification model gives high accuracy but low recall. Analyze what this indicates about the model's performance and prediction errors.	L2	3
21	A medical diagnosis system is designed to identify patients suffering from a serious disease. Which performance measure—Precision or Recall—should be given higher priority? Justify your selection.	L3	3

22	A Random Forest contains 100 decision trees. If 65 trees predict Class 1 and 35 trees predict Class 0, determine the final predicted class using majority voting.	L3	4
23	Evaluate the suitability of Bagging and Boosting for a dataset in which the base decision tree model suffers from high variance. Justify your answer.	L5	4
24	Explain the role of eigenvalues and eigenvectors in Principal Component Analysis.	L2	5
25	Differentiate between AdaBoost and Gradient Boosting based on how they improve the performance of weak learners.	L2	4
26	Explain the difference between Projection and Manifold Learning approaches to dimensionality reduction.	L2	5
27	Analyze how PCA helps overcome the Curse of Dimensionality. Mention any two limitations of PCA.	L4	5
28	Differentiate between Kernel PCA and conventional PCA based on their ability to handle linear and nonlinear data.	L2	5
29	A dataset contains 100 features. After applying PCA, it is reduced to 10 principal components. Calculate the percentage reduction in dimensionality.	L3	5
30	A machine learning problem requires combining predictions from multiple different models, such as Decision Tree, K-NN and SVM. Evaluate whether Voting or Stacking would be more suitable. Give reasons for your choice.	L5	4

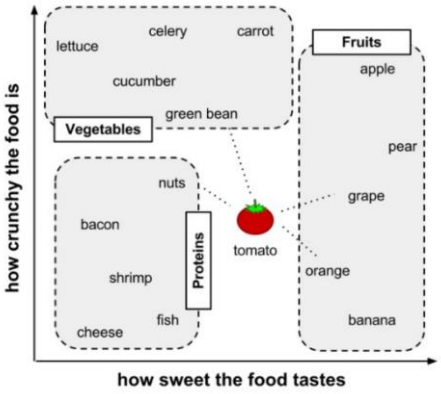
Type of Question- 5 Marks

Q.No	Question	Blooms Level	CO

1	 The network has: • Inputs: (u,v) • Weights: (a,b,c,d,e,f) • Two hidden neurons and one output neuron. • Activation function at each neuron: $$R(x) = \max(0, x)$$ Given: $$u = 2, \quad v = 3$$ $$a = b = c = 1\ , \ \ d = -1$$ $$e = 4, \quad f = -1$$ Find the following derivatives: $\frac{\partial y}{\partial a}$ and $\frac{dy}{df}$	L4	3
2	In Network 1: • Input layer has three neurons: (x_1,x_2,x_3). • First hidden layer has three neurons. • Second hidden layer has two neurons. • Output layer has one neuron. • Every connection shown has weight 1 from the input layer to the first hidden layer, weight 2 from the first to second hidden layer, and weight 3 from the second hidden layer to the output. The connections in the first layer are: • First neuron receives (x_1, x_2). • Second neuron receives (x_1, x_3).	L3	3

	• Third neuron receives (x_1, x_2, x_3). Network 2 directly connects the three inputs to one output neuron, with weights (p, q, r). Given that all three inputs are positive, find the values of (p, q, r) that make both networks equivalent.		
3	An artificial neuron receives three inputs: $$x_1 = 2, x_2 = 1, x_3 = 3$$ The corresponding weights are $$w_1 = 0.5, w_2 = -1 \ \& \ w_3 = 0.2$$ And Bias b=0.1 Calculate : 1. Net input z. 2. Output using the sigmoid activation function $$\sigma(z) = \frac{1}{1 + e^{-z}}$$	L3	3
4	A neuron is represented by: $$y = R(3x - 4)$$ Where $$R(z) = \max(0, z)$$ Calculate 1. Output y at x=1 2. Output y at x=3 3. Derivative $\frac{dy}{dx}$ at both input values	L3	3
5	A neural network has two input neurons, two hidden neurons and one output neuron.	L3	3

	The hidden-layer outputs are: $$h_1 = R(2x_1 + x_2)$$ $$h_2 = R(x_1 - 2x_2)$$ The output is $$y = 4h_1 + 3h_2$$ Given $x_1 = 3$ and $x_2 = 1$ Calculate 1. h_1 2. h_2 3. Final output y		
6	A neural network produces the following predictions: **Sample** | **Actual value** | **Predicted value** 1 | 1 | 0.8 2 | 0 | 0.2 3 | 1 | 0.6 4 | 0 | 0.1 Calculate the mean squared error Q5. A single-neuron network uses sigmoid activation: $y = \sigma(wx + b)$ Given x=2, w=0.5, b=0 Target output t=1 and loss function $L = \frac{1}{2}(t - y)^2$ Learning rate $\eta = 0.1$ Calculate 1. Net input z 2. Predicted Output z 3. Gradient $\frac{\partial L}{\partial w}$	L3	3

	4. Updated weight w_{new}		
7	 **ingredient** / **sweetness** / **crunchiness** / **food type** apple / 10 / 9 / fruit bacon / 1 / 4 / protein banana / 10 / 1 / fruit carrot / 7 / 10 / vegetable celery / 3 / 10 / vegetable cheese / 1 / 1 / protein Using the following dataset find whether tomato is fruit or vegetable or protien.	L3	3
8	Details of ten international cricket games between two teams "Green" and "Blue" are given in Table C. This table consists of matches played on different pitches, across formats along with their winners. The attribute Pitch can take one of two values: spin-friendly (represented as *S*) or pace-friendly (represented as *F*). The attribute Format can take one of two values: one-day match (represented as *O*) or test match (represented as *T*). A cricket organization would like to use the information given in Table C to develop a decision-tree model to predict outcomes of future games between these two teams. To develop such a model, the computed InformationGain(C, Pitch) with respect to the Target is?	L3	3

Match Number	Pitch	Format	Winner (Target)
1	S	T	Green
2	S	T	Blue
3	F	O	Blue
4	S	O	Blue
5	F	T	Green
6	F	O	Blue
7	S	O	Green
8	F	T	Blue
9	F	O	Blue
10	S	O	Green

9	Consider the following dataset:	L4	3

feature1	feature2	output
11.7	183.2	a
12.8	187.6	a
15.3	177.4	a
13.9	198.6	a
17.2	175.3	a
16.8	151.1	b
17.5	171.4	b
23.6	162.8	b
16.9	179.5	b
19.1	173.8	b

Which among the following split-points for the **feature1** would give the best split according to the **information gain** measure?
A. 14.6
B. 16.05
C. 16.85
D. 17.35

For the same dataset, which among the following split-points for *feature2* would give the best split according to the Gini index measure?
Options:
A. 172.6
B. 176.35
C. 178.45

	D. 185.4 Explain your answers with proper calculations.		
10	A Logistic Regression model produces the following predicted probabilities for the positive class: **Sample** | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 **Actual Class** | 1 | 0 | 1 | 1 | 0 | 0 | 1 | 0 **Predicted Probability** | 0.82 | 0.63 | 0.71 | 0.42 | 0.35 | 0.58 | 0.91 | 0.21 . The default classification threshold is 0.5. (a) Convert the predicted probabilities into predicted class labels using a threshold of 0.5. (b) Construct the corresponding confusion matrix. (c) Calculate the following evaluation metrics: i. Accuracy ii. Precision iii. Recall iv. F1-score (d) Now change the classification threshold to 0.7. Calculate the new predicted class labels. (e) Explain how changing the threshold from 0.5 to 0.7 affects false positives and false negatives.	L4	3
11	A Naive Bayes classifier is used for a two-class classification problem with classes y_1 andy_2. The prior probabilities are: $P(y_1) = \frac{1}{3}$, $P(y_2) = \frac{2}{3}$ For a particular feature vector x, the likelihood are P(x| y_1) = $\frac{3}{4}$ and $P(x\|y_2) = \frac{1}{4}$ Calculate the probability of misclassifying x. Round your answer to two decimal places.	L3	3
12	A linear binary classifier is defined as: $f(x) = sign(\omega^T x + b)$ Where $x \in \mathbb{R}^2$ The training dataset consist of: Class 1:	L4	3

	$\left\{\begin{bmatrix}2\\0\end{bmatrix}, \begin{bmatrix}0\\2\end{bmatrix}, \begin{bmatrix}2\\2\end{bmatrix}\right\}$ Class 2: $\left\{\begin{bmatrix}0\\0\end{bmatrix}\right\}$ A hard-margin SVM is trained on this dataset to obtain w and b. Find the number of support vectors and margin.		
13	PCA is applied to reduce the dimensionality of a feature space from 100 to 10 for a classification problem. What is the angle θ between the first and tenth principal components?	L2	5
14	A dataset contains n observations, each represented by a 100-dimensional vector: $D = \{x^{(1)}, x^{(2)}, \ldots\ldots, x^{(n)}\}$ Where $x^{(i)} \in \mathbb{R}^{100}$ It is given that $\sum_{i=1}^{n} x^{(i)} = 0$ The covariance matrix of this dataset has Eigen values: $\lambda_i = 100^{2-i}, \ 1 \leq i \leq 100$ Let u be the unit direction of maximum variance, such that: $u^T u = 1$ Calculate the following quantity $\frac{1}{n}\sum_{i=1}^{n}\left(u^T x^{(i)}\right)^2$	L3	5
15	Define clustering in machine learning. Explain the different types of clustering methods with suitable examples.	L2	2
16	Explain how fuzzy clustering can be applied to segment customers based on their purchasing behaviour. Discuss the role of membership values in assigning customers to different clusters.	L2	2
17	Define classification in machine learning. List and briefly describe the major classification algorithms, including Logistic Regression, Decision Tree, K-NN, SVM and Naive Bayes.	L2	3
18	Analyze the differences between Decision Tree and Logistic Regression classifiers in terms of their working principles, decision boundaries, interpretability and suitability for different types of datasets.	L4	3
19	A healthcare organization wants to develop a machine learning model to identify patients suffering from a serious disease. Evaluate the suitability of K-NN, Logistic Regression, Decision Tree and Naive Bayes for this application. Also justify which performance measures should be	L5	3

	prioritized while evaluating the model.		
20	Define ensemble learning. List and briefly explain the major ensemble learning techniques used in machine learning.	L2	4
21	Explain how a Random Forest classifier can be developed for predicting whether a customer will purchase a product. Describe the roles of bootstrap sampling, random feature selection and majority voting in the classification process.	L2	4
22	Explain the working principle of Principal Component Analysis (PCA). Discuss the roles of eigenvalues, eigenvectors and explained variance ratio in identifying principal components.	L2	5
23	Explain the working principles of Bagging and Boosting. How do these techniques improve the performance of individual machine learning models?	L2	4
24	Define Machine Learning. Explain its major objectives and list any four real-world applications of Machine Learning.	L2	1
25	Define the Machine Learning Life Cycle. List and briefly describe the major stages involved in developing a Machine Learning model.	L2	1
26	Define supervised learning and unsupervised learning. List their major types with suitable examples.	L2	1
27	Define Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL) and Data Science (DS). Explain their basic meaning.	L2	1
28	Explain the differences between batch learning and online learning. Discuss their working principles and suitable applications.	L2	1
29	Differentiate between instance-based learning and model-based learning with suitable examples.	L2	1
30	Explain the importance of data visualization in Machine Learning. Discuss any five commonly used data visualization techniques and their purposes.	L2	1
31	A company wants to predict house prices based on area, location and number of rooms. Identify the appropriate type of Machine Learning and explain how the Machine Learning Life Cycle can be applied to solve this problem.	L3	1
32	A dataset contains features with significantly different numerical ranges, such as age (18–60), salary (10,000–1,00,000) and experience (0–30). Explain how normalization can be applied to bring these features to a comparable scale.	L2	1
33	A Machine Learning model performs very well on training data but gives poor results on unseen test data. Analyze the possible reasons for this behaviour and suggest suitable methods to improve its generalization performance.	L4	1
34	A business organization wants to analyze customer purchasing behaviour but does not have predefined output labels. Evaluate whether supervised or unsupervised learning is more appropriate for this problem. Justify your answer.	L3	1
35	Evaluate the suitability of batch learning and online learning for an application involving continuous data generation, such as stock market	L5	1

	monitoring or real-time fraud detection. Justify your choice.		
36	A dataset contains missing values, duplicate records, outliers and features with different scales. Evaluate the importance of different data preprocessing techniques and recommend a suitable preprocessing strategy.	L5	1
37	Explain with the help of example why we need to split the data into training, testing and validation with proper mathematical expressions.	L2	1
38	In the context of machine learning briefly explain no free lunch theorem.	L2	1
39	Briefly describe how artificial neural network is inspired from biological neural network.	L2	3
40	Derive an expression for weight updation using back propagation when two neurons are connected and the activation function is sigmoid.	L3	3
41	Explain how KNN serves as a promising algorithm for application with instances. What happens when the dimensions are increased.	L4	3
42	A confusion matrix shows 80 correctly classified positive instances, 70 correctly classified negative instances, 10 false positives and 20 false negatives. Calculate the accuracy of the classifier.	L3	3
43	A dataset has three principal component eigenvalues: 6, 3 and 1. Calculate the percentage of total variance explained by the first principal component.	L3	5
44	Differentiate between overfitting and underfitting in Machine Learning. Explain their effect on model performance.	L2	1
45	A dataset contains many features, making model training difficult. Evaluate how PCA can help reduce dimensionality and improve model efficiency.	L3	5
46	Write a Python program using pandas to read a CSV dataset and display its first five rows, number of rows and columns, and basic statistical information.	L3	1
47	Study the following code and answer the questions: from sklearn.metrics import accuracy_score y_actual = [1, 0, 1, 1, 0] y_pred = [1, 0, 0, 1, 0] print(accuracy_score(y_actual, y_pred)) a) Write the output. b) How many predictions are correct? c) Explain the meaning of the obtained accuracy.	L3	3
48	Observe the following code and answer the questions:	L3	4

	from sklearn.ensemble import RandomForestClassifier X = [[1], [2], [3], [4], [5], [6]] y = [0, 0, 0, 1, 1, 1] model = RandomForestClassifier(n_estimators=10, random_state=42) model.fit(X, y) print(model.predict([[2]])) print(model.predict([[5]])) a) What is the purpose of `n_estimators=10`? b) What does `fit()` do? c) What does `predict()` return for a new input?		
49	Study the following code and answer the questions: from sklearn.decomposition import PCA import numpy as np X = np.array([[2, 4, 6], [4, 6, 8], [6, 8, 10]]) pca = PCA(n_components=2) X_new = pca.fit_transform(X) print(X_new.shape) print(pca.n_components_) a) What is the purpose of `n_components=2`? b) Write the expected output of `X_new.shape`. c) How many features are present in the transformed dataset?	L3	5
50	What is the difference between idi and temporal data. How you should split each one of them for model training.	L2	1

Type of Question- Read and Observer --15 Marks

<table>
<tr><th>Q.No</th><th>Question</th><th>Blooms Level</th><th>CO</th></tr>
<tr><td>1</td><td>A college wants to develop a Machine Learning system to predict whether a student will pass or fail based on attendance, internal marks, and assignment performance.

(a) Observe the given data and identify the input features and target variable. (5 Marks)

Student Attendance (%) Internal Marks Assignment Score Result
A 85 18 9 Pass
B 45 8 4 Fail
C 75 16 8 Pass
D 40 7 3 Fail

(b) Identify the type of Machine Learning used in this problem. Give a reason for your answer. (5 Marks)

(c) Write any five major steps involved in the Machine Learning life cycle for developing this system. (5 Marks)</td><td>L2</td><td>1</td></tr>
<tr><td>2</td><td>A dataset contains the following information about students:

Student Age Attendance (%) Marks
A 20 80 75
B 21 85
C 19 60 55
D 22 100 95

(a) Observe the table and identify the missing value. Suggest one suitable method to handle it. (5 Marks)

(b) Which visualization technique would you use to represent the attendance of students? Give a reason for your choice. (5 Marks)

(c) Explain the need for data normalization and data augmentation in Machine Learning with one example of each. (5 Marks)</td><td>L2</td><td>1</td></tr>
<tr><td>3</td><td>A shopping mall wants to group its customers based on their annual income and spending score. The following customer information is</td><td>L2</td><td>2</td></tr>
</table>

	available: **Customer Annual Income (₹ in Lakhs) Spending Score** A 2 20 B 3 25 C 8 80 D 9 85 E 4 30 F 10 90 (a) Observe the data and suggest how customers can be grouped into clusters based on similar characteristics. (5 Marks) (b) Identify whether this problem requires supervised or unsupervised learning. Justify your answer. (5 Marks) (c) Name and briefly explain any two clustering algorithms suitable for this problem. (5 Marks)		
4	A Machine Learning model is developed to predict whether an email is Spam or Not Spam. After testing, the following confusion matrix is obtained: **Actual / Predicted Spam Not Spam** Spam 40 10 Not Spam 5 45 (a) Identify True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN) from the given confusion matrix. (5 Marks) (b) Calculate the classification accuracy of the model using the given confusion matrix. (5 Marks) (c) Name any two classification algorithms that can be used for spam detection and briefly explain their working principles. (5 Marks)	L3	3
5	A company develops three different Machine Learning models to predict whether a customer will purchase a product. The predictions of the three models are given below: **Customer Model 1 Model 2 Model 3**	L3	4

	A Yes Yes No B No Yes No C Yes Yes Yes D No No Yes (a) Using majority voting, determine the final prediction for each customer. (5 Marks) (b) Explain the basic idea of ensemble learning and state one advantage of combining multiple models. (5 Marks) (c) Differentiate between Bagging and Boosting with one suitable example of each. (5 Marks)			
6	A dataset contains 100 features related to customer purchasing behaviour. Many of these features may contain similar or redundant information. A company wants to reduce the number of features while retaining important information. (a) What is dimensionality reduction? Explain the curse of dimensionality in simple terms. (5 Marks) (b) Observe the following explained variance ratios obtained after applying PCA: **Principal Component Explained Variance Ratio** PC1 50% PC2 25% PC3 15% PC4 10% Calculate the total variance retained by the first two principal components. (5 Marks) (c) Briefly explain the role of PCA in dimensionality reduction. Also define PAC learning and VC dimension. (5 Marks)		L2	5